1. Lesson 2: The Fourier transform in L^1 and L^2

1.1. **Fourier trasnfrom in L^1.** In this course we shall use the following definition of Fourier transform.

Definition 1.1. Let $f \in L^1(\mathbb{R}^n)$ and we set

$$\hat{f}(\xi) := \int_{\mathbb{R}^n} f(y) e^{-2i\pi\xi\cdot y} dy, \qquad \forall \xi \in \mathbb{R}^n, \tag{1.1}$$

with $\xi \cdot y = \sum_{j=1}^n \xi_j y_j$. The function $\hat{f} : \mathbb{R}^n \to \mathbb{C}$ is called Fourier transform of f.

Remark 1.2. At times, we will also use the notation $\mathcal{F}(\dots)$ instead of $\hat{f}(\dots)$.

Remark 1.3. The Fourier transform (1.1) is well defined. Indeed

$$\begin{aligned} |f(y)e^{-2i\pi\xi\cdot y}| &= |f(y)||\cos(2\pi i\xi\cdot y) - i\sin(2\pi i\xi\cdot y)| = |f(y)|\sqrt{\cos^2(2\pi i\xi\cdot y) + \sin^2(2\pi i\xi\cdot y)} \\ &= |f(y)| \in L^1(\mathbb{R}^n) \end{aligned}$$

Theorem 1.4 (properties of the Fourier transform)**.** *Let $f \in L^1(\mathbb{R}^n)$. Then*

i) $\hat{f} \in C^0_*(\mathbb{R}^n)$ (i.e. $\hat{f}$ is continous and goes to zero at infinity) with

$$\|\hat{f}\|_{C^0(\mathbb{R}^n)} \le \|f\|_{L^1(\mathbb{R}^n)};$$

in particular the operator Fourier transform:

$$\begin{aligned} \mathcal{F} \quad : \quad & L^1(\mathbb{R}^n) \to C^0(\mathbb{R}^n) \\ & f \mapsto \hat{f}, \end{aligned}$$

is linear and continous.

ii) If also $g \in L^1(\mathbb{R}^n)$ then

$$\widehat{f * g} = \hat{f} \cdot \hat{g}^1$$

iii) If we consider an integer $k \ge 1$ and $x^\alpha f \in L^1(\mathbb{R}^n)$ for any multi-index [2] with $|\alpha| \le k$ then $\hat{f} \in C^k(\mathbb{R}^n)$ and

$$\partial^\alpha \hat{f} = \mathcal{F}((-2\pi i x)^\alpha f).$$

iv) If we consider an integer $k \ge 1$ such that $f \in C^{k-1}(\mathbb{R}^n)$, $\partial^\alpha f \in C^0_*(\mathbb{R}^n)$ for $|\alpha| \le k-1$ and furthermore $\partial^\alpha f \in L^1(\mathbb{R}^n)$ for $|\alpha| \le k$, then

$$\widehat{\partial^\alpha f}(\xi) = (2\pi i\xi)^\alpha \hat{f}(\xi).$$

iv) If we consider $g \in L^1(\mathbb{R}^n)$, then

$$\int_{\mathbb{R}^n} f(y)\hat{g}(y)dy = \int_{\mathbb{R}^n} \hat{g}(y)f(y)dy.$$

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[1] The convolution of two function f and g is defined by

$$(f * g)(x) = \int_{\mathbb{R}^n} f(y)g(y-x)dy$$

.

[2] It means: $\alpha = (\alpha_1, \alpha_2, ..., \alpha_n)$ with $\alpha_j \in \mathbb{N}$, $|\alpha| = \alpha_1 + \alpha_2 + ...\alpha_n$,

$$x^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} ... x_n^{\alpha_n}$$

$$\partial^\alpha f = \frac{\partial^{|\alpha| f}}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} ... \partial x_n^{\alpha_n}}$$

v) Let $\eta \in \mathbb{R}^n$ be fixed. Then we have

$$\mathcal{F}(f(x)e^{2\pi i x\cdot\eta}) = \hat{f}(\xi - \eta).$$

vi) Let $\eta \in \mathbb{R}^n$ be fixed. Then we have

$$\mathcal{F}(f(. + \eta))(\xi) = e^{2\pi i \xi\cdot\eta}\hat{f}(\xi).$$

Remark 1.5. The first property means that the Fourier transform in L^1 does not map its domain to itself.

Example 1.6. *Compute the Fourier transform of*

$$f(x) = e^{-|x|}, \qquad x \in \mathbb{R}.$$

We have to compute

$$\hat{f}(\xi) = \int_{\mathbb{R}} e^{-|y|}e^{-2\pi i\xi y}dy.$$

By using the elementary properties of integrals and the Euler identity we get

$$\begin{aligned}\int_{\mathbb{R}} e^{-|y|}e^{-2\pi i\xi y}dy &= \int_{\mathbb{R}} e^{-|y|}\cos(2\pi\xi y)dy - i\int_{\mathbb{R}} e^{-|y|}\sin(2\pi\xi y)dy \\ &= 2\int_0^\infty e^{-y}\cos(2\pi\xi y)dy\end{aligned}$$

By integrating two times by parts we have

$$2\int_0^\infty e^{-y}\cos(2\pi\xi y)dy = \frac{2}{1+4\pi^2\xi^2}.$$

Thus we get

$$\hat{f}(\xi) = \frac{2}{1+4\pi^2\xi^2}. \tag{1.2}$$

Example 1.7. *Compute the Fourier transform of the Gaussian function*

$$f(x) = e^{-x^2}, \qquad x \in \mathbb{R}.$$

We start observing that

$$f'(x) = -2xe^{-x^2}.$$

Thus f satisfies the ODE

$$f'(x) = -2xf(x).$$

We apply the Fourier transform to the ODE, by the properties iii) and iv) of Theorem 1.4 we have

$$(2\pi i\xi)\hat{f}(\xi) = -2\frac{1}{-2\pi i}(\hat{f})'(\xi)$$

$$(\hat{f})'(\xi) = (-2\pi^2\xi)\hat{f}(\xi).$$

This is an ODE with separation of variables and the solution is given by

$$\hat{f}(\xi) = ce^{-x^2\xi^2},$$

with

$$c = \hat{f}(0) = \int_{\mathbb{R}} e^{-x^2}dx = \sqrt{\pi}.$$

Thus we have

$$\hat{f}(\xi) = \sqrt{\pi}e^{-x^2\xi^2}.$$

Theorem 1.8 (Inversion theorem)**.** *Let $f \in L^1(\mathbb{R}^n)$. If also $\hat{f} \in L^1(\mathbb{R}^n)$ we have that*

$$f(x) = \int_{\mathbb{R}^n} \hat{f}(\xi) e^{2\pi i \xi \cdot x} d\xi, \qquad a.e \quad x \in \mathbb{R}^n. \tag{1.3}$$

In symbols we can write

$$f(x) = \widehat{(\hat{f})}(-x) \qquad a.e \quad x \in \mathbb{R}^n.$$

Example 1.9. *By* (1.2) *and the inversion theorem of the Fourier transform we deduce that*

$$\mathcal{F}\left(\frac{2a}{a^2 + 4\pi^2 x^2}\right)(\xi) = e^{-a|\xi|},$$

and so for $a = 2\pi$ we have

$$\mathcal{F}\left(\frac{1}{1+x^2}\right)(\xi) = \pi e^{-2\pi|\xi|}.$$

Remark 1.10 (Limitations of the inversion theorem)**.** By Theorem 1.4, we know that for any $f \in L^1(\mathbb{R}^n)$, the Fourier transform $\hat{f}$ is continuous and vanishes at infinity. However, this property does not guarantee that $\hat{f}$ is integrable, which is a necessary condition for the validity of the Fourier inversion theorem. Therefore, it is not always true that an integrable function can be "reconstructed" from its Fourier transform via the identity: $f(x) = \widehat{(\hat{f})}(-x)$.

Example 1.11. *Compute the Fourier transform in $\mathbb{R}$ of the following function*

$$f(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

By the definition of the Fourier transform we have

$$\begin{aligned} \hat{f}(\xi) &= \int_0^{+\infty} e^{-x} e^{-2\pi i x \xi} dx \\ &= \lim_{b \to +\infty} \left[\frac{e^{-x(1+2\pi i \xi)}}{-(1+2\pi i \xi)} \right]_0^b \\ &= \frac{1}{1+2\pi i \xi} \\ &= \frac{1}{1+4\pi^2\xi^2} - \frac{2\pi \xi i}{1+4\pi^2\xi^2}. \end{aligned}$$

We observe that

$$|\hat{f}(\xi)| = \frac{1}{\sqrt{1+4\pi^2\xi^2}} \notin L^1(\mathbb{R}).$$

This is an example of a function that is not integrable, and as a result, the inversion theorem does not apply.

In this lecture, we briefly explored the theory of the Fourier transform on the space $L^1(\mathbb{R}^n)$, which is sufficient for certain applications but not all. We identified compelling reasons to extend the theory to the space $L^2(\mathbb{R}^n)$ and to a suitable space of distributions. These extensions enable us to compute the Fourier transform of non-integrable functions such as 1, $\sin(\omega x)$, and others.

1.2. **Fourier transform in L^2.** In order to consider the Fourier transform in L^2 we need to define the space of rapidly decreasing functions.

Definition 1.12. We say that a function $f : \mathbb{R}^n \to \mathbb{R}$ (or $\mathbb{C}$) is a rapidly decreasing function if $f \in C^\infty(\mathbb{R}^n)$ and for any integer $k \geq 0$ and multi-index $\alpha \geq 0$ we have

$$\lim_{|x|\to\infty} |x|^\alpha \partial^\alpha f(x) = 0.$$

We denote this space by $\mathcal{S}(\mathbb{R}^n)$.

In words, the function f and any of its derivatives (of any order) tend to zero at infinity faster than any negative power of $|x|$.

Example 1.13. *The function $f(x) = e^{-|x|^2}$ belong to $\mathcal{S}(\mathbb{R}^n)$. Indeed $\lim_{|x|\to\infty} |x|^k \partial^\alpha e^{-|x|^2} = 0$, $\forall \alpha, k$.*

The space $\mathcal{S}(\mathbb{R}^n)$ is a vectorial space but it is not a normed space. However, the following important theorem of density in $L^p(\mathbb{R}^n)$ is valid.

Theorem 1.14. *The space $\mathcal{S}(\mathbb{R}^n)$ is dense in $L^p(\mathbb{R}^n)$ for any $p \in [1, \infty)$. This means that for all $f \in L^p(\mathbb{R}^n)$, $p \in [1, \infty)$, there exists $\{f_n\} \subset \mathcal{S}(\mathbb{R}^n)$ such that*

$$\|f_n - f\|_{L^p(\mathbb{R}^n)} \to 0, \qquad n \to \infty.$$

Since every function in $\mathcal{S}(\mathbb{R}^n)$ belongs, a fortiori, to $L^1(\mathbb{R}^n)$, the Fourier transform is well-defined on the space $\mathcal{S}(\mathbb{R}^n)$.

Theorem 1.15 (Fourier transform on the space $\mathcal{S}(\mathbb{R}^n)$)**.** *(i) For any $f \in \mathcal{S}(\mathbb{R}^n)$ we have $\hat{f} \in \mathcal{S}(\mathbb{R}^n)$. Moreover the inversion formula of the Fourier transform is valid.*

(ii) For any f, $g \in \mathcal{S}(\mathbb{R}^n)$ we have

$$\int_{\mathbb{R}^n} f(x)\bar{g}(x)dx = \int_{\mathbb{R}^n} \hat{f}(\xi)\bar{\hat{g}}(\xi)d\xi,$$

where we denote by $\bar{g}$ the complex conjugate.

(iii) For any $f \in \mathcal{S}(\mathbb{R}^n)$ we have

$$\|f\|_{L^2(\mathbb{R}^n)} = \|\hat{f}\|_{L^2(\mathbb{R}^n)}.$$

Remark 1.16. From the first point of the above theorem we have that

$$\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n), \qquad \mathcal{F}^{-1} : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n).$$

Since the Fourier transform maps the space $\mathcal{S}(\mathbb{R}^n)$ to itself, it preserves the L^2-norm. Furthermore, the fact that $\mathcal{S}(\mathbb{R}^n)$ is dense in L^2 allows the Fourier transform to be defined on L^2.

Theorem 1.17 (Independence from approximating sequence)**.** *Let $f \in L^2(\mathbb{R}^n)$ and let $\{f_n\} \subset \mathcal{S}(\mathbb{R}^n)$ such that $\|f_n - f\|_{L^2} \to 0$, $n \to \infty$. Then the sequence $\{\hat{f}_n(\xi)\}$ converges to a function that belongs to L^2 and such a function does not depend on the specific approximating sequence.*

The above theorem means that if we consider $f \in L^2$ we can approximate the function with different sequences of $\mathcal{S}(\mathbb{R}^n)$. Let us consider

$$g_n \to f, \quad h_n \to f \quad \text{in} \quad L^2.$$

The functions $\{\hat{g}_n(\xi)\}$, $\{\hat{h}_n(\xi)\}$ converge to the same function $\hat{f}(\xi)$ in L^2:

$$\hat{g}_n(\xi) \to \hat{f}(\xi), \quad \hat{h}_n(\xi) \to \hat{f}(\xi) \quad \text{in} \quad L^2.$$

The approximation does not depend on the approximating sequences g_n or h_n that we are considering.

Definition 1.18 (Fourier transform in $L^2(\mathbb{R}^n)$)**.** Let $f \in L^2(\mathbb{R}^n)$ and $\{f_k\} \subset \mathcal{S}(\mathbb{R}^n)$ such that $\|f_k - f\|_{L^2} \to 0$, $k \to \infty$. Then $\hat{f}$ is defined as the limit in $L^2(\mathbb{R}^n)$ of $\{\hat{f}_k\} \subset \mathcal{S}(\mathbb{R}^n)$.

Now, we list the properties of the Fourier transform in $L^2(\mathbb{R}^n)$.

Theorem 1.19. *The operator*

$$\mathcal{F} : L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$$

is linear and continuos, and satisfies the following identities:

$$(f, g)_{L^2} = (\hat{f}, \hat{g})_{L^2},$$

$$\|f\|_{L^2(\mathbb{R}^n)} = \|\mathcal{F}(f)\|_{L^2(\mathbb{R}^n)},$$

for all f, $g \in L^2(\mathbb{R}^n)$. Moreover $\mathcal{F}$ is a bijection form $L^2(\mathbb{R}^n)$ to itself.

Remark 1.20. It is also valid the inversion formula

$$f(x) = \int_{\mathbb{R}^n} \hat{f}(\xi) e^{2\pi i \xi \cdot x} d\xi = \mathcal{F}(\hat{f}(\xi))(-x), \qquad \forall f \in L^2(\mathbb{R}^n).$$